UNIVERSITAT POLITÈCNICA DE CATALUNYA
BARCELONATECH
Departament de Matemàtiques

3D modelling of confined gastruloid aggregates

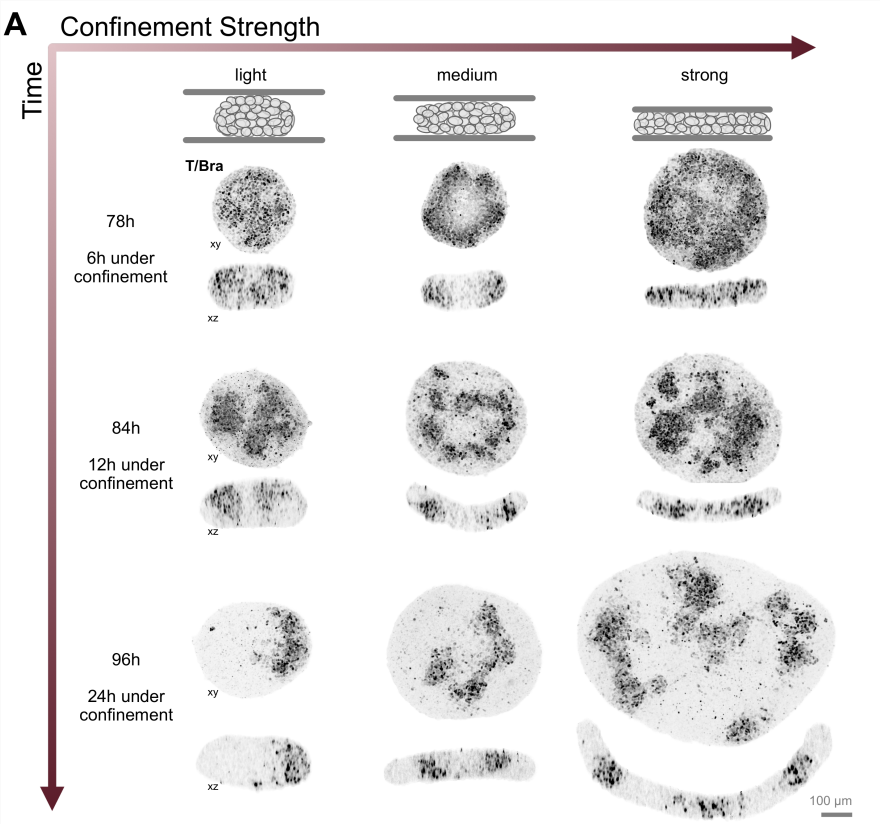
An agent-based model of T/Bra patterning under confinement

Víctor Villegas-Morral

Joint work with D. Oriola and V. Weichselberger

Lab meeting
November 2026

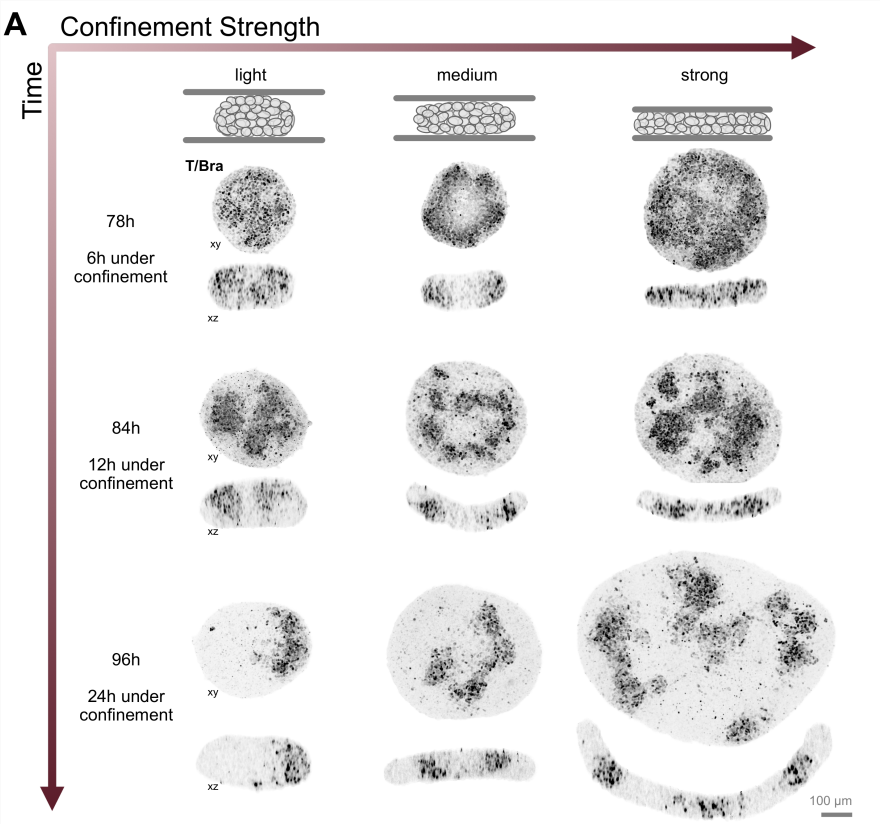
Gastruloids under confinement



Gastruloids are squeezed between two plates at 72 h, right before T/Bra polarises, and read at 96 h.

Images: V. Weichselberger.

Gastruloids under confinement

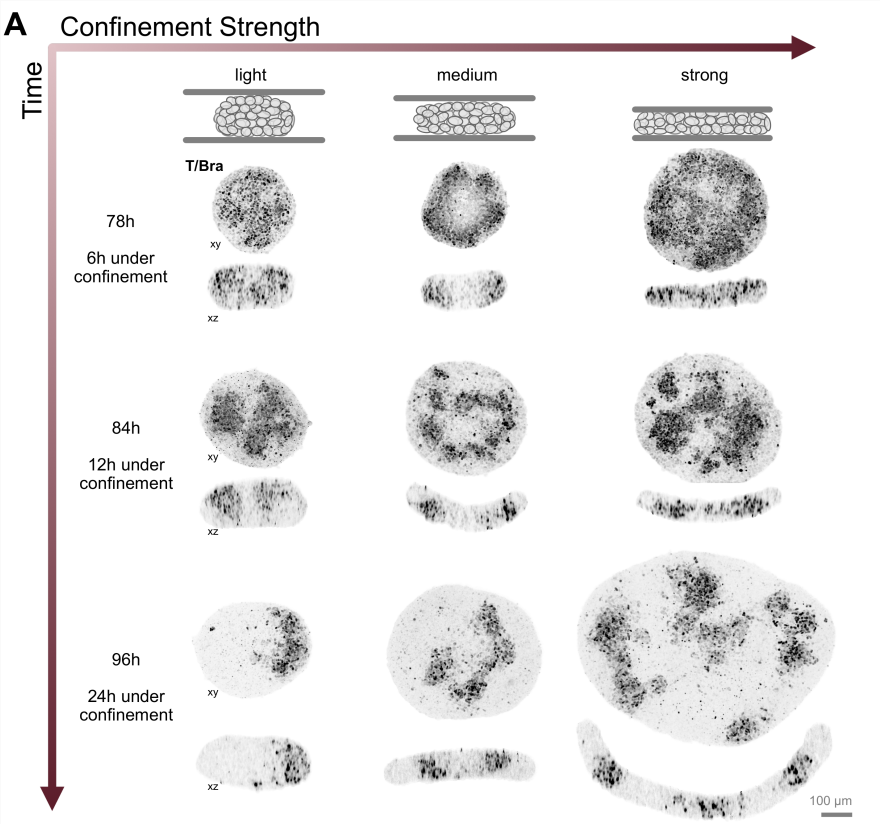


Gastruloids are squeezed between two plates at 72 h, right before T/Bra polarises, and read at 96 h.

- **Light**: T/Bra polarises, as without confinement.

Images: V. Weichselberger.

Gastruloids under confinement

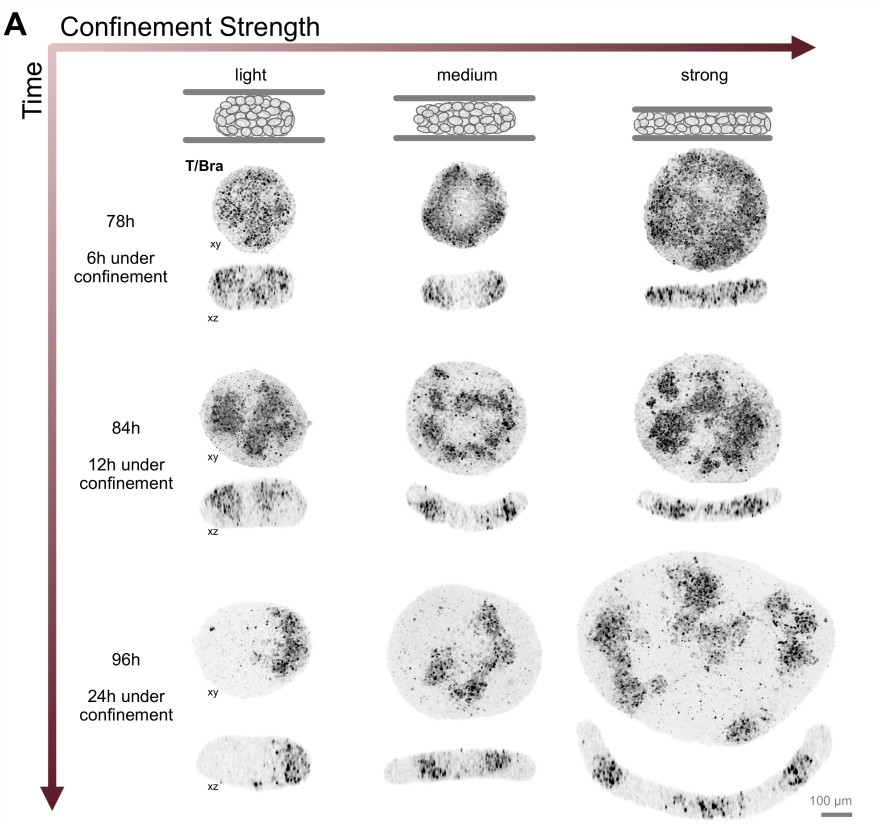


Gastruloids are squeezed between two plates at 72 h, right before T/Bra polarises, and read at 96 h.

- **Light**: T/Bra polarises, as without confinement.
- **Medium**: T/Bra stays in the centre, sometimes as a ring.

Images: V. Weichselberger.

Gastruloids under confinement

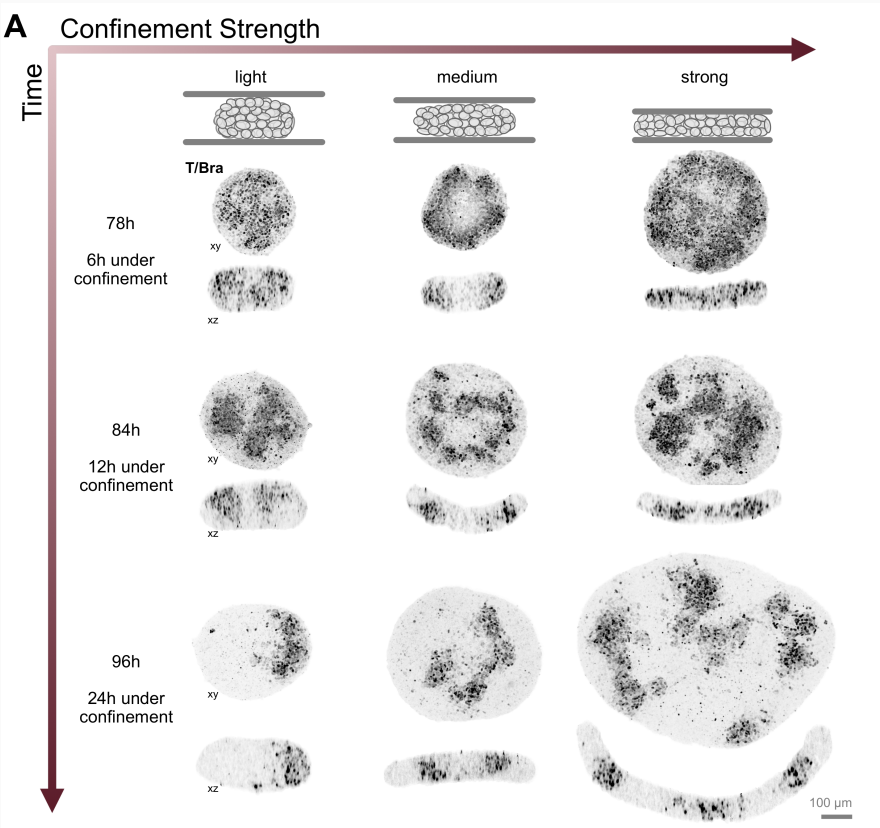


Gastruloids are squeezed between two plates at 72 h, right before T/Bra polarises, and read at 96 h.

- **Light**: T/Bra polarises, as without confinement.
- **Medium**: T/Bra stays in the centre, sometimes as a ring.
- **Strong**: T/Bra breaks into clusters.

Images: V. Weichselberger.

Gastruloids under confinement



Gastruloids are squeezed between two plates at 72 h, right before T/Bra polarises, and read at 96 h.

- **Light**: T/Bra polarises, as without confinement.
- **Medium**: T/Bra stays in the centre, sometimes as a ring.
- **Strong**: T/Bra breaks into clusters.

The larger the gastruloid, the thicker the slab that still gives clusters.

Images: V. Weichselberger.

Goal

A single model, with a single set of parameters, that reproduces

- polarisation without confinement,
- a central cluster under medium confinement,
- clusters under strong confinement,
- and the fusion of aggregates of the same and of different cell types.

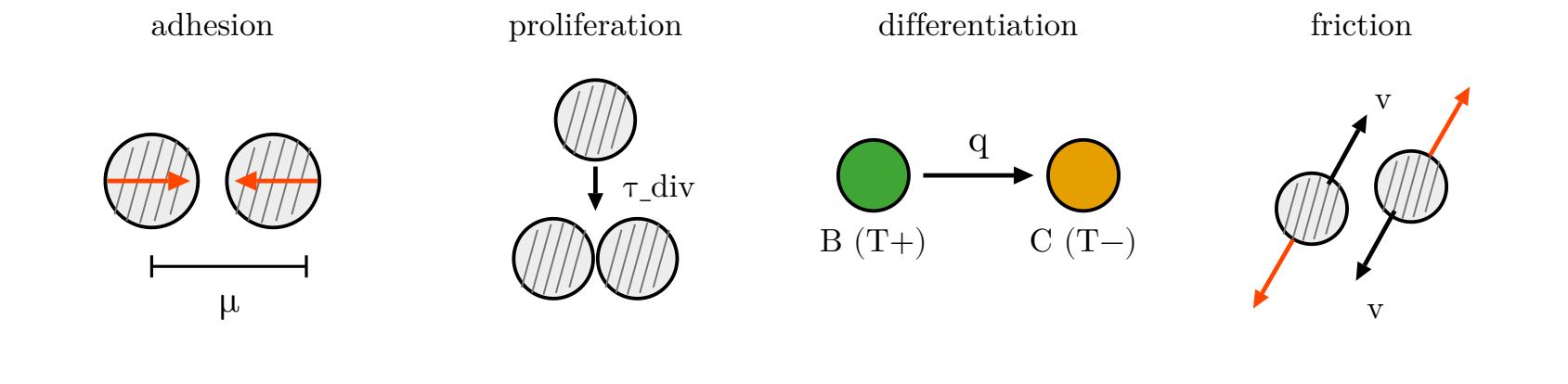
Goal

A single model, with a single set of parameters, that reproduces

- polarisation without confinement,
- a central cluster under medium confinement,
- clusters under strong confinement,
- and the fusion of aggregates of the same and of different cell types.

What sets the outcome: the number of cells, the thickness of the slab, or both?

The model in one slide



- Each cell is a sphere. Neighbours **attract and repel** each other, with strengths that depend on the state of both cells.
- Cells **divide** after a random time (**B** every 20 h, **C** every 50 h).
- **B** (T/Bra^{+}) cells **differentiate** into **C** (T/Bra^{-}) at a constant rate q.
- Two rigid **plates**, with some in-plane **friction** on the cells that touch them.

D Oriola et al. (2026). 'Collective fate decisions and cell rearrangements underlie gastruloid symmetry breaking'. *bioRxiv*. DOI: 10.1101/2024.12.16.628776. URL: https://www.biorxiv.org/content/early/2026/02/24/2024.12.16.628776

Mechanics

Overdamped dynamics

$$\dot{\mathbf{x}}_i = \mathbf{v}_i \,, \qquad \mathbf{v}_i = \frac{1}{\lambda n_i}\left(\sum_j \mathbf{F}_{ij} + \alpha \sum_j \mathbf{v}_j\right)$$

$$\mathbf{F}_{ij} = f(d_{ij})\,\hat{\mathbf{r}}_{ij} \times \begin{cases} F_{\text{rep}}^{(s_i,s_j)} & d_{ij} \le 2r \\ F_{\text{att}}^{(s_i,s_j)} & 2r < d_{ij} < 2\mu_{s_i s_j} r \end{cases} \qquad f(d) = \left(\frac{2r}{d} - 1\right)\left(\frac{2\mu r}{d} - 1\right)$$

Mechanics

Overdamped dynamics

$$\dot{\mathbf{x}}_i = \mathbf{v}_i\,, \qquad \mathbf{v}_i = \frac{1}{\lambda n_i}\left(\sum_j \mathbf{F}_{ij} + \alpha \sum_j \mathbf{v}_j\right)$$

$$\mathbf{F}_{ij} = f(d_{ij})\,\hat{\mathbf{r}}_{ij} \times \begin{cases} F_{\text{rep}}^{(s_i,s_j)} & d_{ij} \le 2r \\ F_{\text{att}}^{(s_i,s_j)} & 2r < d_{ij} < 2\mu_{s_i s_j} r \end{cases} \qquad f(d) = \left(\frac{2r}{d} - 1\right)\left(\frac{2\mu r}{d} - 1\right)$$

Plates

Rigid plates at $z = 0$ and $z = L$, with $L = 2.1\,r \times$ (number of layers). Cells touching a plate feel an extra in-plane drag λ_{wall}.

Mechanics

Overdamped dynamics

$$\dot{\mathbf{x}}_i = \mathbf{v}_i\,, \qquad \mathbf{v}_i = \frac{1}{\lambda n_i}\left(\sum_j \mathbf{F}_{ij} + \alpha \sum_j \mathbf{v}_j\right)$$

$$\mathbf{F}_{ij} = f(d_{ij})\,\hat{\mathbf{r}}_{ij} \times \begin{cases} F_{\text{rep}}^{(s_i,s_j)} & d_{ij} \le 2r \\ F_{\text{att}}^{(s_i,s_j)} & 2r < d_{ij} < 2\mu_{s_i s_j} r \end{cases} \qquad f(d) = \left(\frac{2r}{d} - 1\right)\left(\frac{2\mu r}{d} - 1\right)$$

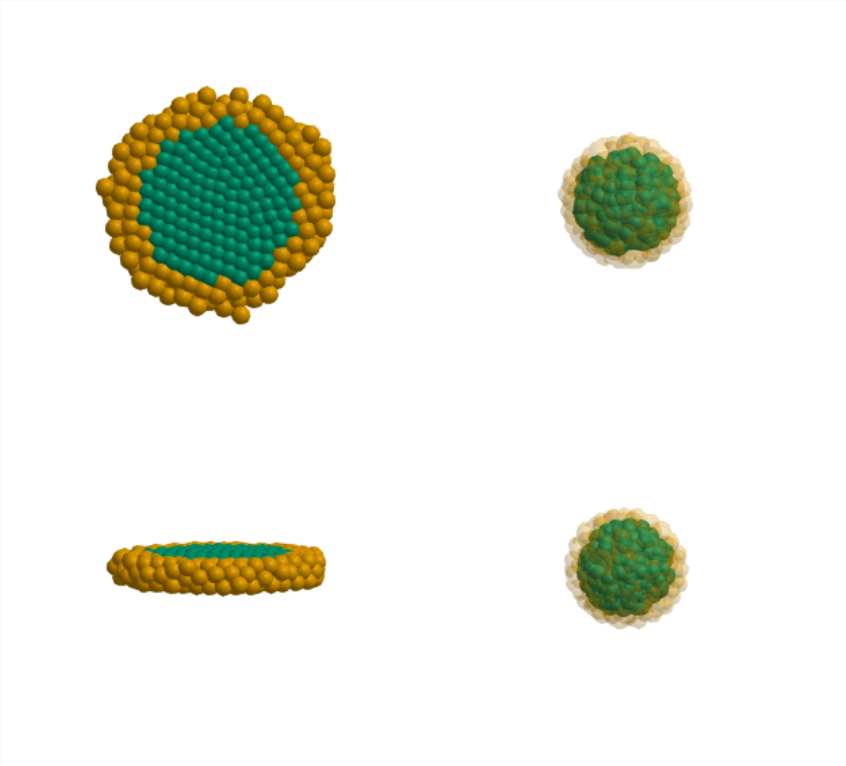
Plates

Rigid plates at $z = 0$ and $z = L$, with $L = 2.1\,r \times$ (number of layers). Cells touching a plate feel an extra in-plane drag λ_{wall}.

Simplifications

No *A* state: at squeezing only **B** and **C** cells remain. No noise: cells move only because neighbours divide and push, and because adhesion changes with differentiation.

Initial condition and readout



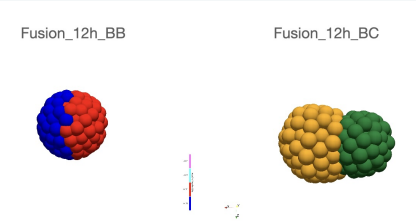
- An already squeezed aggregate: grown between the plates and relaxed.
- A central core of **B** cells with 60% of the cells; the rest are **C**.
- The outcome is read when 20% of the cells are **B**, about 18 h later. In the experiments T/Bra$^+$ falls from 60% to 20% between 72 and 96 h.

Left: 2 layers, top view cut at the mid-plane. Right: unconfined, **C** translucent. Both $N_0 = 600$.

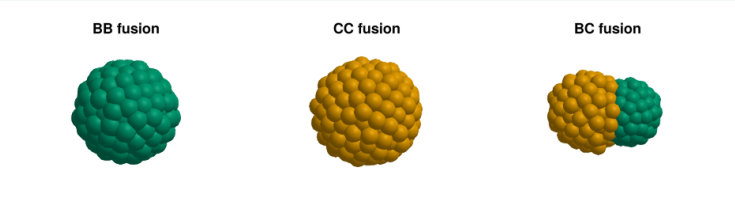
Sanity checks

Fusion

BB and **CC** doublets fuse into one sphere; **BC** doublets glue but do not round up.



Expected

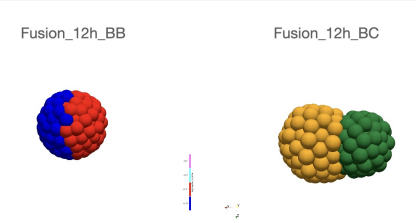


Model, after 24 h

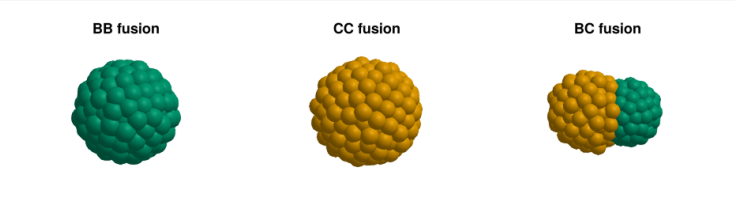
Sanity checks

Fusion

BB and **CC** doublets fuse into one sphere; **BC** doublets glue but do not round up.



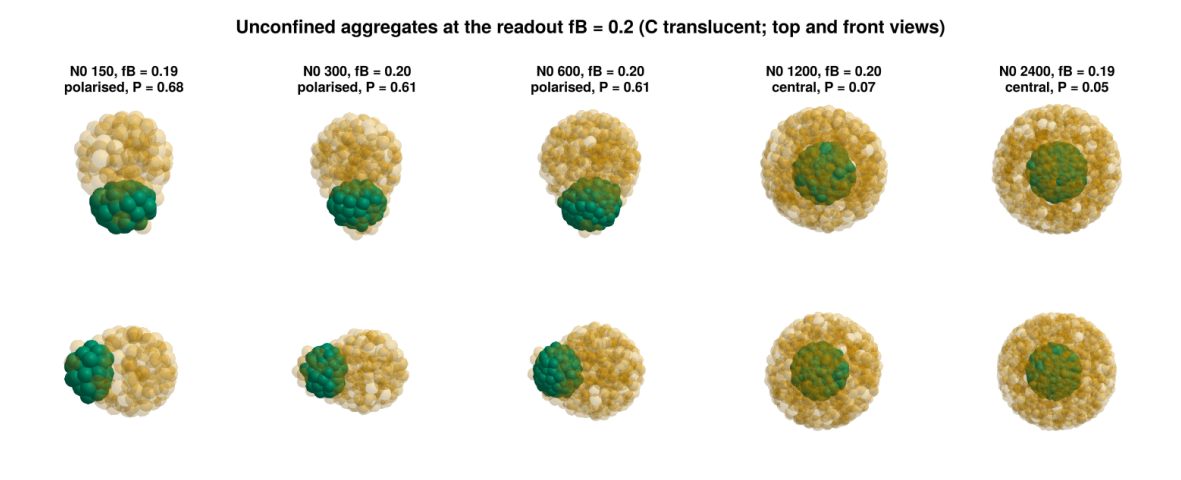
Expected



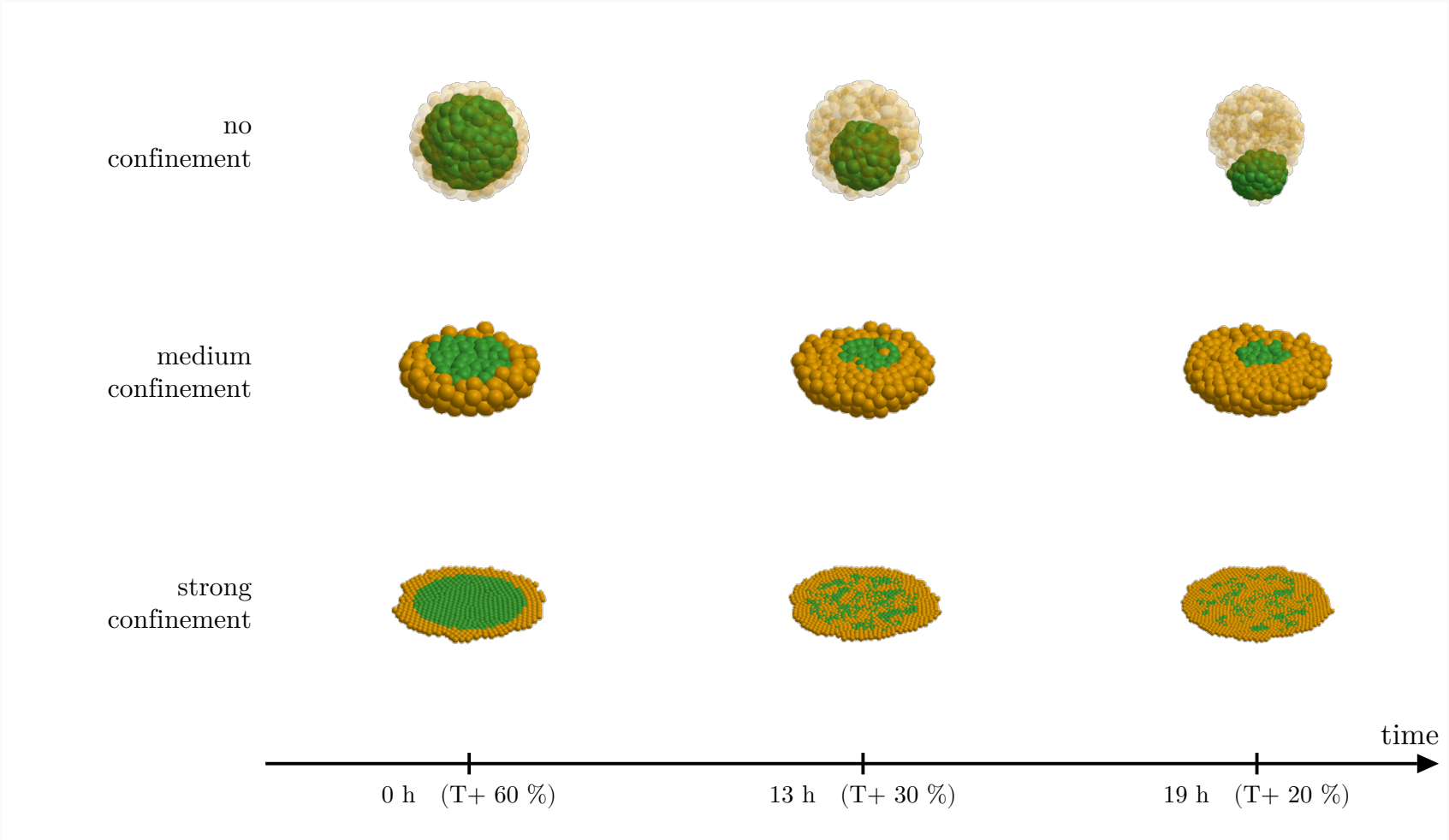
Model, after 24 h

Polarisation

Unconfined aggregates of up to $N_0 = 600$ cells polarise before **B** runs out.

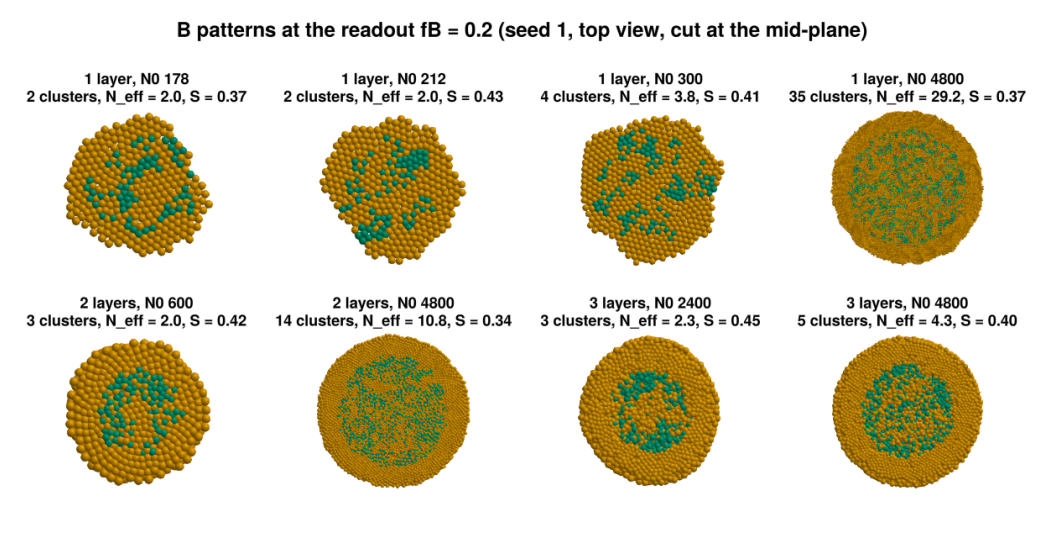


The three behaviours



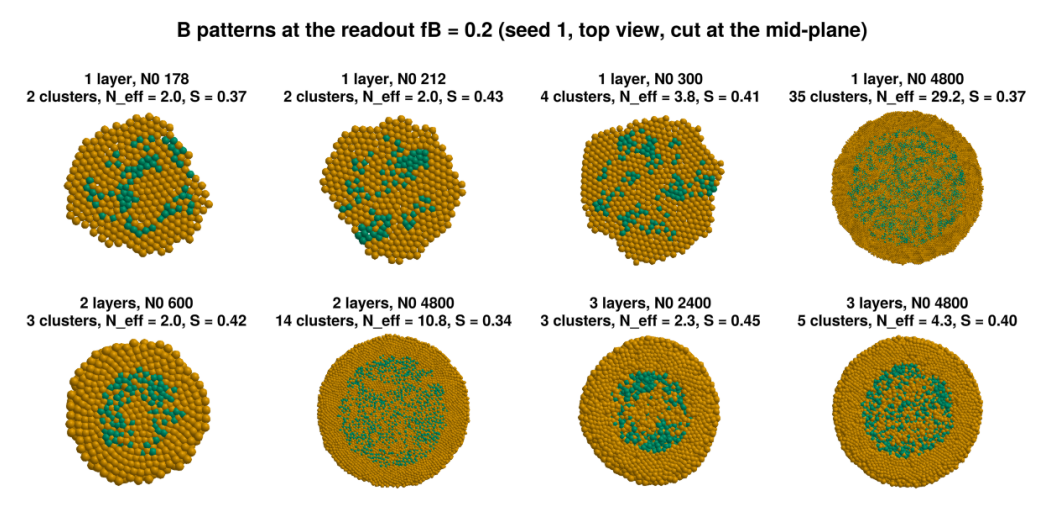
The same aggregate (N_0 = 600) with no, medium (4 layers) and strong (1 layer) confinement.

Patterns



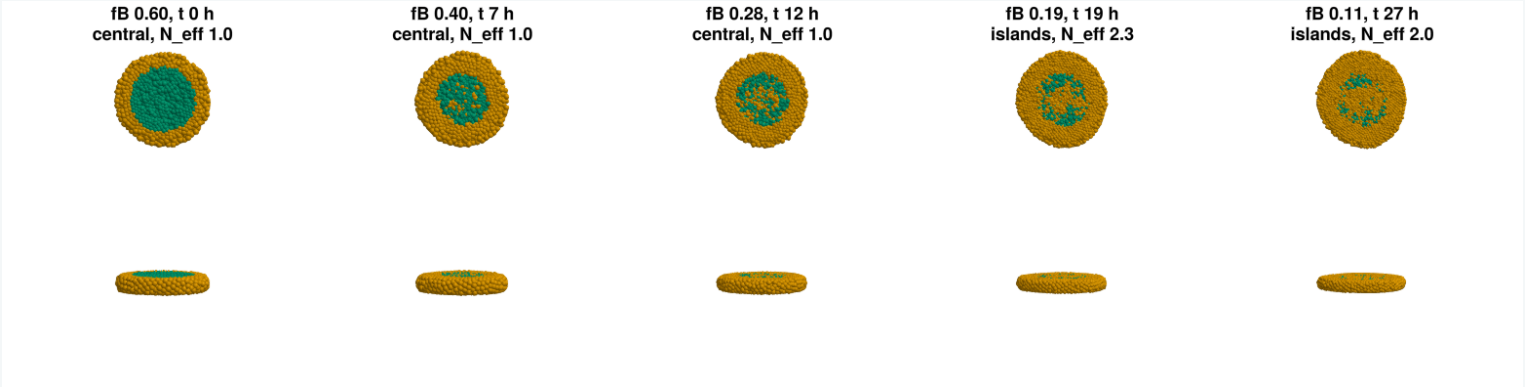
From one to three layers, when 20% of the cells are **B**. Top view cut at the mid-plane.

Patterns



From one to three layers, when 20% of the cells are **B**. Top view cut at the mid-plane.

In thick slabs the core first hollows into a **ring**, which then breaks into islands.



What makes thick slabs pattern

- In one or two layers, new **C** cells inside the **B** core cannot get out and **split the core**.
- From three layers on, cells pass over and under each other: a cohesive core **squeezes the C cells out** and stays as one cluster.

What makes thick slabs pattern

- In one or two layers, new **C** cells inside the **B** core cannot get out and **split the core**.
- From three layers on, cells pass over and under each other: a cohesive core **squeezes the C cells out** and stays as one cluster.

The B–B adhesion energy

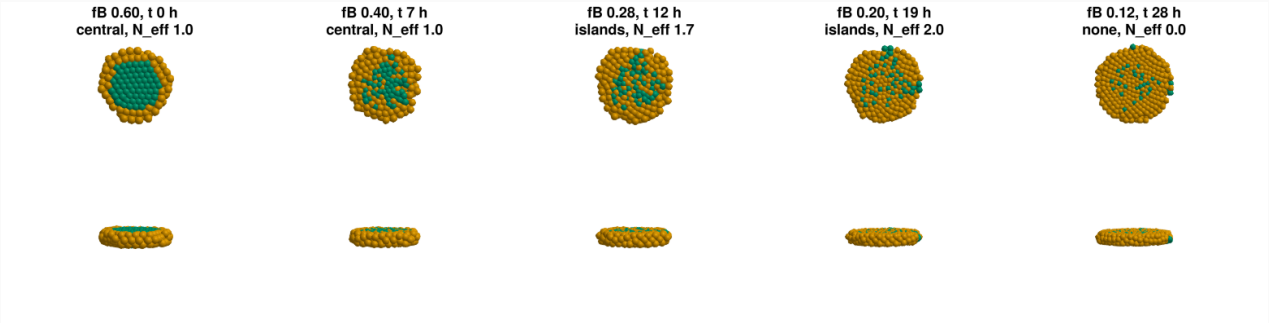
- Shortening the **B**–**B** attraction range, $\mu_{BB} : 2 \rightarrow 1.9$, lowers W_{BB} from 1.02 to 0.79, still above $W_{CC} = 0.65$. Large cores in thick slabs now open into rings.
- With $\mu_{BB} = 1.85$, $W_{BB} \approx W_{CC}$: **B** is no longer more cohesive than **C** and **dissolves** into it.

What makes thick slabs pattern

- In one or two layers, new **C** cells inside the **B** core cannot get out and **split the core**.
- From three layers on, cells pass over and under each other: a cohesive core **squeezes the C cells out** and stays as one cluster.

The B–B adhesion energy

- Shortening the **B**–**B** attraction range, $\mu_{BB} : 2 \rightarrow 1.9$, lowers W_{BB} from 1.02 to 0.79, still above $W_{CC} = 0.65$. Large cores in thick slabs now open into rings.
- With $\mu_{BB} = 1.85$, $W_{BB} \approx W_{CC}$: **B** is no longer more cohesive than **C** and **dissolves** into it.



Reading the outcome as in the experiments

From the positions and states when 20% of the cells are **B**:

- **Clusters**: the experiments' density-based clustering of **B** cells (radius 13 μm), and the effective number of clusters $N_{\text{eff}} = 1/\sum_k p_k^2$.
- **Polarisation**: $P = |\bar{\mathbf{x}}_B - \bar{\mathbf{x}}|/R$, with R half the length along the longest axis.

Reading the outcome as in the experiments

From the positions and states when 20% of the cells are **B**:

- **Clusters**: the experiments' density-based clustering of **B** cells (radius 13 μm), and the effective number of clusters $N_{\text{eff}} = 1/\sum_k p_k^2$.
- **Polarisation**: $P = |\bar{\mathbf{x}}_B - \bar{\mathbf{x}}|/R$, with R half the length along the longest axis.

Outcome

1. **Polarised** if $P \geq 0.3$;
2. **Clusters** if $N_{\text{eff}} \geq 1.5$;
3. **Central cluster** otherwise (an unbroken ring counts as central).

Reading the outcome as in the experiments

From the positions and states when 20% of the cells are **B**:

- **Clusters**: the experiments' density-based clustering of **B** cells (radius 13 μm), and the effective number of clusters $N_{\text{eff}} = 1/\sum_k p_k^2$.
- **Polarisation**: $P = |\bar{\mathbf{x}}_B - \bar{\mathbf{x}}|/R$, with R half the length along the longest axis.

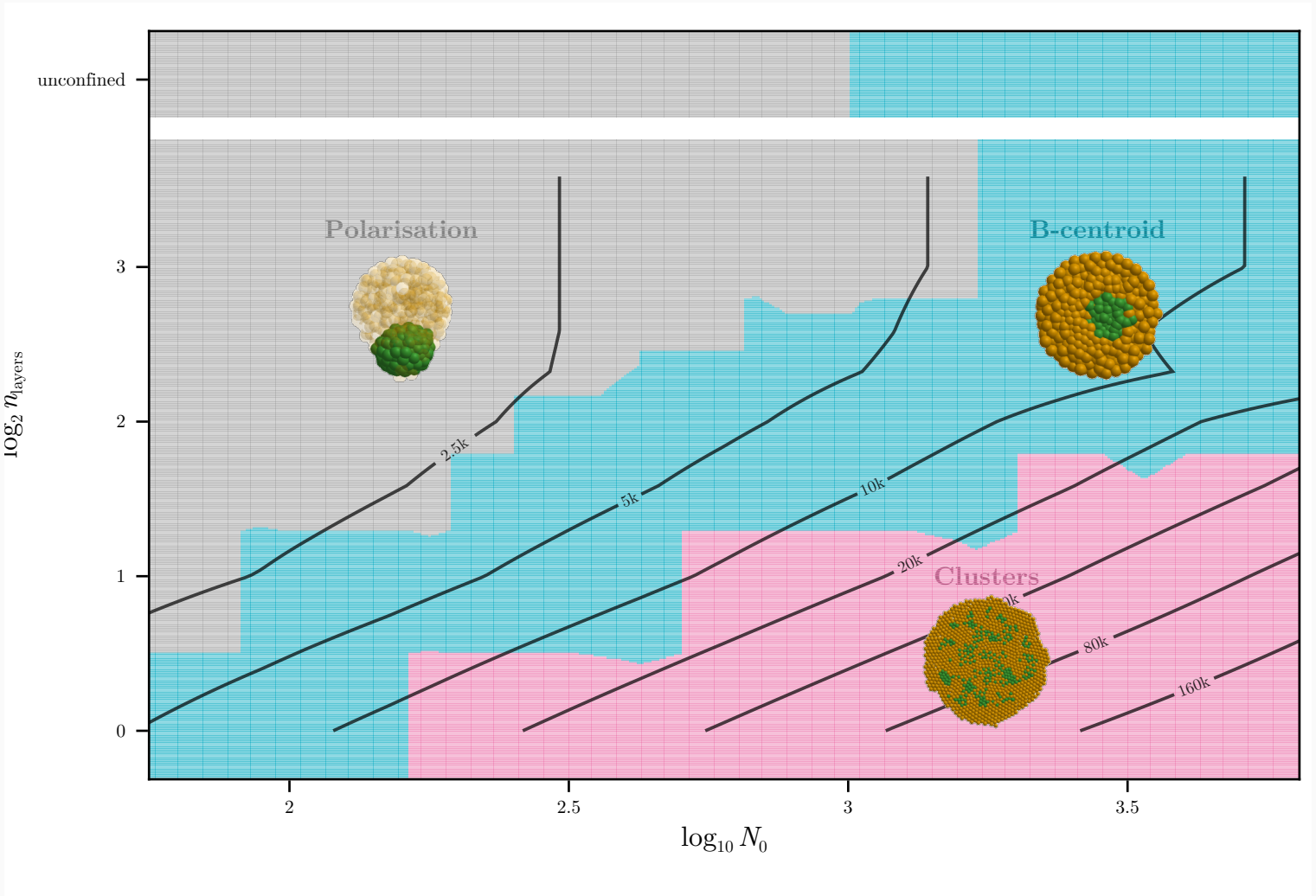
Outcome

1. **Polarised** if $P \geq 0.3$;
2. **Clusters** if $N_{\text{eff}} \geq 1.5$;
3. **Central cluster** otherwise (an unbroken ring counts as central).

Changes for our smaller aggregates (10 to 100 times fewer cells)

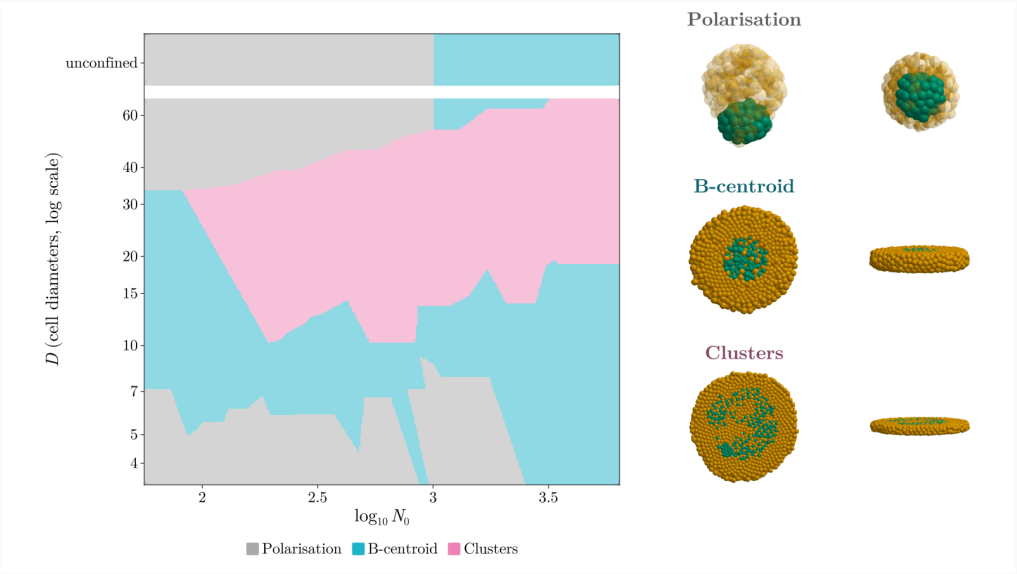
Clusters of at least 10 cells instead of 20, and a core cell needs 50% of **B** neighbours instead of 40%.

Phase space



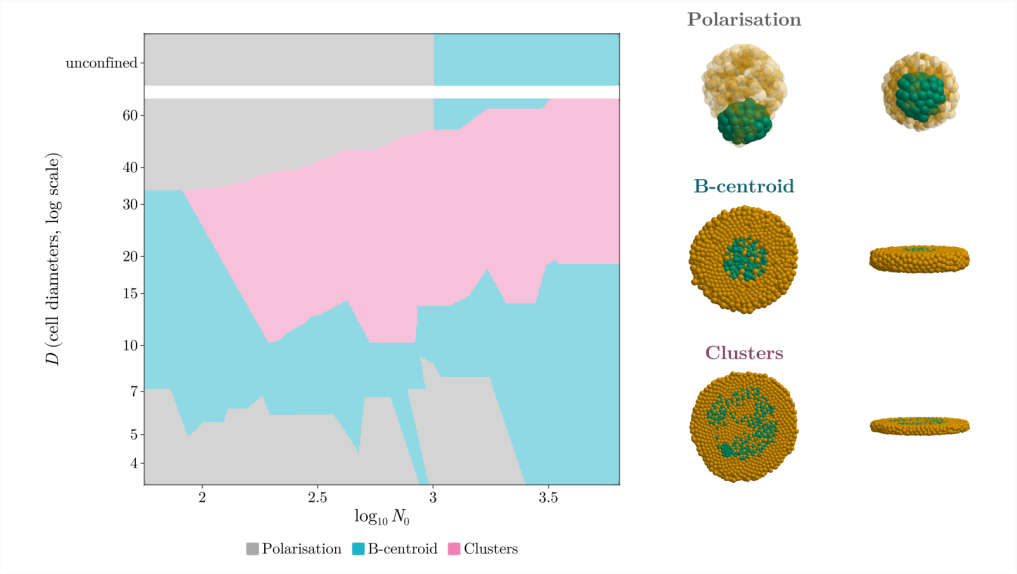
Number of cells at squeezing against the number of layers, 2 or 3 simulations per point. Lines: area of the pancake at squeezing (μm^2).

The width of the pancake sets the threshold



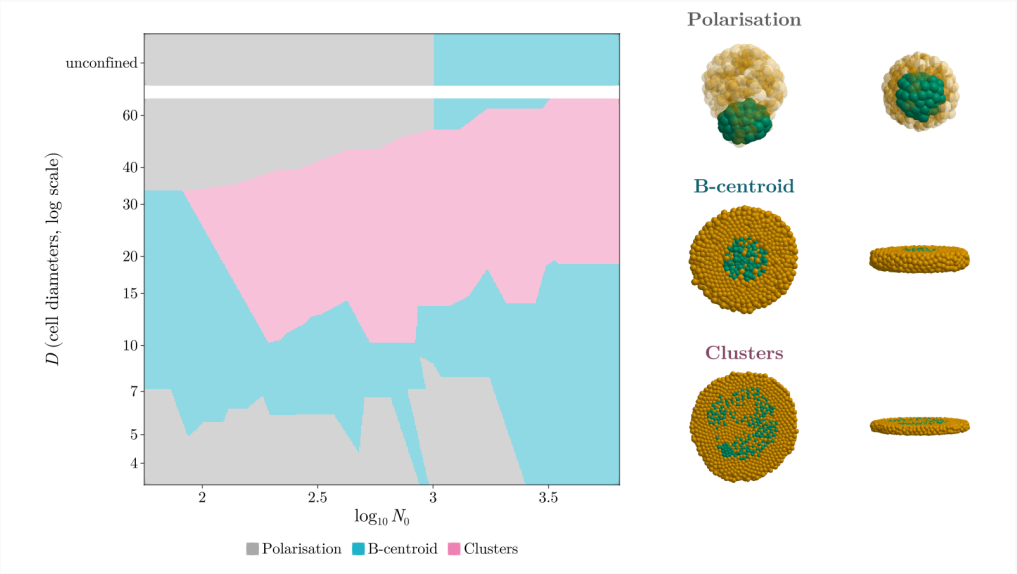
- The threshold in N_0 grows fast with thickness: from $\approx$ 160 cells at one layer to $\approx$ 4000 at four.

The width of the pancake sets the threshold



- The threshold in N_0 grows fast with thickness: from $\approx$ 160 cells at one layer to $\approx$ 4000 at four.
- Against the **diameter** of the pancake it barely moves: patterns appear at 8 to 17 cell diameters, roughly 80 to 170 μm.

The width of the pancake sets the threshold



- The threshold in N_0 grows fast with thickness: from $\approx$ 160 cells at one layer to $\approx$ 4000 at four.
- Against the **diameter** of the pancake it barely moves: patterns appear at 8 to 17 cell diameters, roughly 80 to 170 μm.
- A thicker slab simply needs more cells to reach the same width.

Differentiation is faster than mixing

The Damköhler number compares differentiation with cell mixing over a length ℓ,

$$\mathrm{Da} = \frac{q\,\ell^2}{D}.$$

Differentiation is faster than mixing

The Damköhler number compares differentiation with cell mixing over a length ℓ,

$$\mathrm{Da} = \frac{q\,\ell^2}{D}.$$

- Measured in growing monolayers: $D \approx 1.5\ \mu\mathrm{m}^2/\mathrm{h}$ for **B** and **C** alike.
- A cell needs about 17 h to move one diameter, longer than a **B** cell lives ($\approx$ 14 h).

Differentiation is faster than mixing

The Damköhler number compares differentiation with cell mixing over a length ℓ,

$$\mathrm{Da} = \frac{q\,\ell^2}{D}.$$

- Measured in growing monolayers: $D \approx 1.5\ \mu\mathrm{m}^2/\mathrm{h}$ for **B** and **C** alike.
- A cell needs about 17 h to move one diameter, longer than a **B** cell lives ($\approx$ 14 h).
- Over the pancake, $\mathrm{Da} \approx 0.015\,A/\mu\mathrm{m}^2$: the iso-area lines of the phase space are lines of constant Da, and the pattern threshold sits at $\mathrm{Da} \approx 200$.

Differentiation is faster than mixing

The Damköhler number compares differentiation with cell mixing over a length ℓ,

$$\mathrm{Da} = \frac{q\,\ell^2}{D}.$$

- Measured in growing monolayers: $D \approx 1.5\ \mu\mathrm{m}^2/\mathrm{h}$ for **B** and **C** alike.
- A cell needs about 17 h to move one diameter, longer than a **B** cell lives ($\approx$ 14 h).
- Over the pancake, $\mathrm{Da} \approx 0.015\,A/\mu\mathrm{m}^2$: the iso-area lines of the phase space are lines of constant Da, and the pattern threshold sits at $\mathrm{Da} \approx 200$.

The **B** pattern is set where the cells already are.

Conclusions and open points

Conclusions

- One parameter set gives polarisation, a central cluster, rings and islands, and the right fusion behaviour.
- The cohesion of the **B** tissue decides whether thick slabs pattern.
- The pattern threshold is set by the width of the pancake, not by its thickness.

Conclusions and open points

Conclusions

- One parameter set gives polarisation, a central cluster, rings and islands, and the right fusion behaviour.
- The cohesion of the **B** tissue decides whether thick slabs pattern.
- The pattern threshold is set by the width of the pancake, not by its thickness.

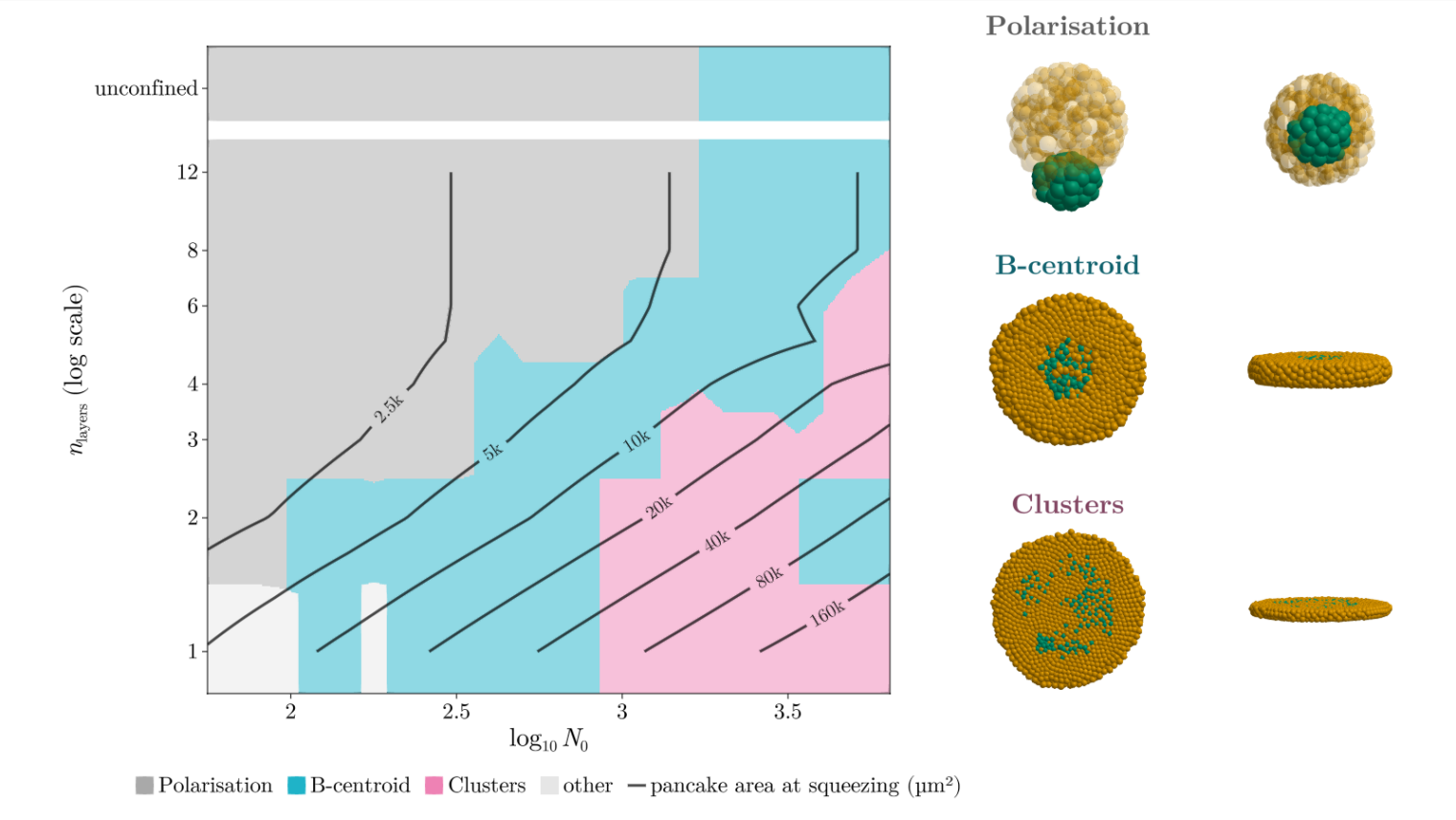
Open points

- Large unconfined aggregates neither pattern nor polarise before **B** runs out.
- μ_{BB} = 1.9 is close to the point where **B** dissolves (1.85).
- Are the diameter at squeezing and the cell diffusion coefficient measured in the experiments? They would test the threshold and Da directly.

Thank you for your attention!

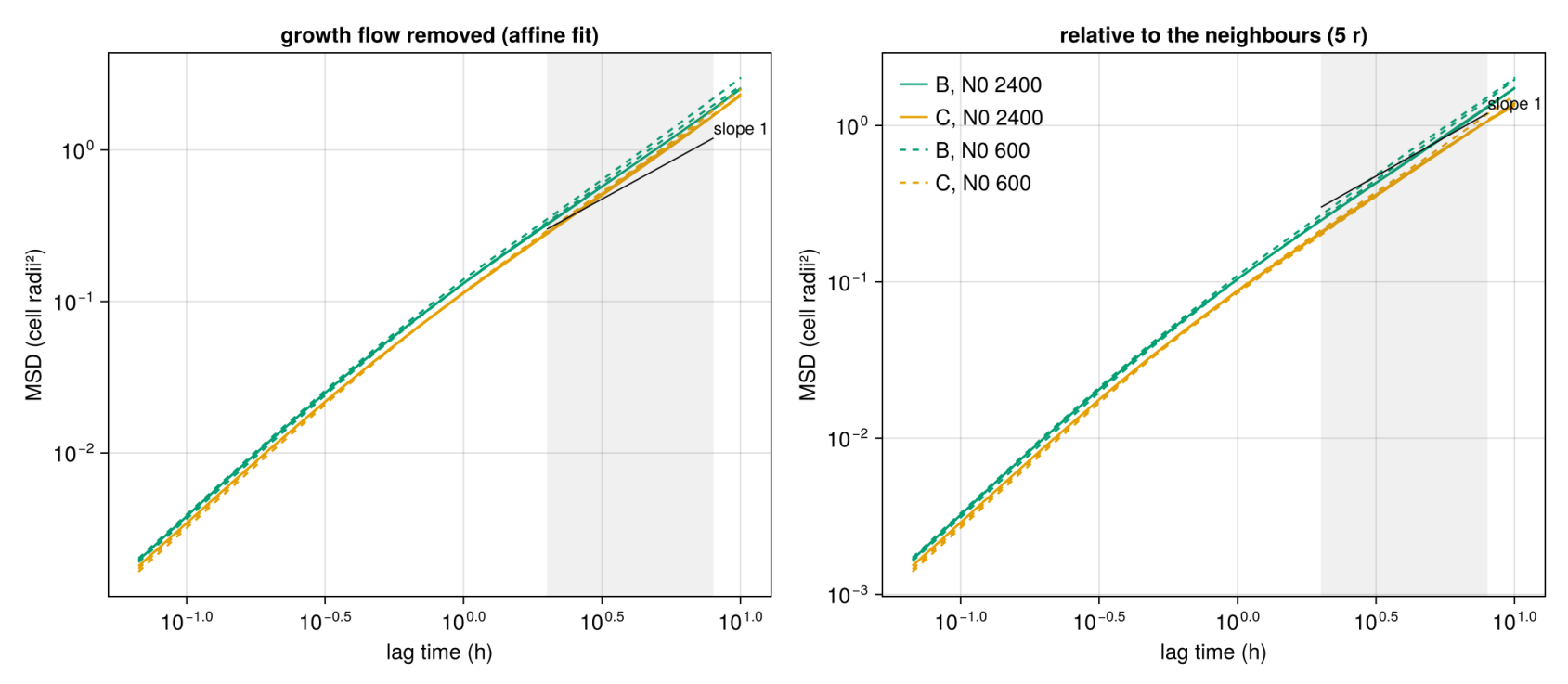
Gràcies per la vostra atenció!

Phase space read later



The same simulations when 12% of the cells are **B**: rings break into islands, larger aggregates polarise late, and the smallest monolayer patterns shrink back to a central cluster.

Mean squared displacement



B (green) and **C** (yellow) cells in growing monolayers (N_0 = 600 dashed, 2400 solid). Left: growth flow removed with an affine fit; right: relative to the neighbours within 5r. Shaded: the fitted lag times.

Parameters

Symbol	Description	Value
r	Radius	$1 = 5\ \mu$m
λ, α	Friction, cell-cell friction	1, 0.9
τ^B_{div}, τ^C_{div}	Division times	300, 750 (20 h, 50 h)
q	**B** $\rightarrow$ **C** rate	0.004667
μ_{BB}, μ_{BC}, μ_{CC}	Force range factors	1.9, 2, 2.14
F^{BB}_{rep}, F^{BC}_{rep}, F^{CC}_{rep}	Repulsion	4.5, 3, 6
F^{BB}_{att}, F^{BC}_{att}, F^{CC}_{att}	Attraction	6.4, 1.6, 3
λ_{wall}	Plate friction	1.5
L	Plate gap	$2.1\,r\times$ layers
T_0	Time unit	4 min